

# The slow experiment: searching for self-amplifying handedness at nature’s speed

## Aim

Test whether simple early-Earth chemistry in water can **amplify a small excess of one hand** when given months to years instead of days. The spec sheet (README Part 6) shows that speed is never the problem for nature, yet lab searches usually stop after days.

## Hypothesis (stated before any data)

At least one prebiotic, water-based peptide-forming system shows **seed-following growth of enantiomeric excess** (ee) at meteoritic concentrations ( $\leq 1\text{ mM}$ ), with an effective rate  $k_2 \geq 3 \times 10^{-6}\text{ /M/s}$ . That rate is slow enough to be missed by short experiments but fast enough to matter in a 1 mM pond over 100 years (the “pond floor”).

*Revised 2026-09-30 after advice from J. Dworkin (NASA Goddard).* The first version ran at 0.1 M with an assumed 0.2% ee precision. Meteoritic amino acids reach at most a few hundred nmol/g of rock, which is  $\leq \sim 1\text{ mM}$  in parent-body water. Published ee uncertainties are  $\pm 0.01\text{--}1.5\%$  (GC-MS) and  $1.2\text{--}7.2\%$  (LC-MS), and 2.6% from repeat measurements (Glavin & Dworkin 2009).

## Why a *seeded* design

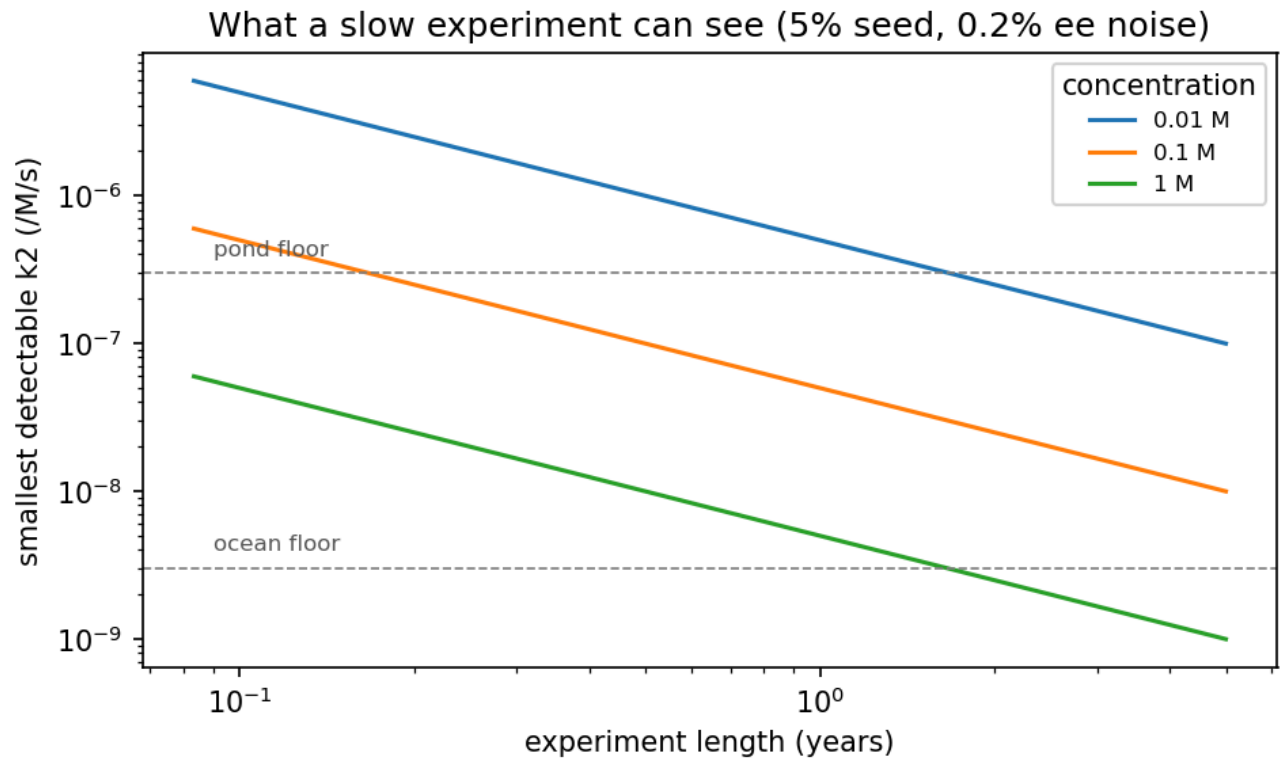
A reaction that only copies itself keeps ee constant, because both hands grow equally. ee *grows* only if the hands also **suppress each other**. So watching ee grow from a small seed tests self-copying and mutual suppression together (spec R1 + R2), which is exactly the missing combination. Seeding is also far faster than waiting for symmetry to break from pure 50/50. From 50/50, noise in a 50  $\mu\text{L}$ , 1 M vial would need about 18 e-foldings of growth to become visible, versus under 1 from a 5% seed.

## How long and how concentrated (`design.py`, `review.py` R5)

Smallest detectable rate (/M/s) with a **20% seed**, similar to the largest isovaline excesses in Murchison. Each time point is measured 9 times, so the noise is  $\sigma/3$ :

| concentration       | ee noise per measurement | 1 month            | 6 months           | 1 year               | 3 years                   |
|---------------------|--------------------------|--------------------|--------------------|----------------------|---------------------------|
| 1 mM (realistic)    | 2.6%                     | $6 \times 10^{-5}$ | $1 \times 10^{-5}$ | $5 \times 10^{-6}$   | $2 \times 10^{-6}$ ✓ pond |
| 1 mM                | 1%                       | $3 \times 10^{-5}$ | $4 \times 10^{-6}$ | $2 \times 10^{-6}$ ✓ | $7 \times 10^{-7}$        |
| 10 mM (accelerated) | 2.6%                     | $6 \times 10^{-6}$ | $1 \times 10^{-6}$ | $5 \times 10^{-7}$   | $2 \times 10^{-7}$        |
| 0.1 M (accelerated) | 2.6%                     | $6 \times 10^{-7}$ | $1 \times 10^{-7}$ | $5 \times 10^{-8}$   | $2 \times 10^{-8}$        |

Nature’s floors at realistic concentrations: **pond**  $3\times10^{-6}$  (1 mM, 100 yr); **ocean**  $3\times10^{-9}$  (1  $\mu$ M,  $10^8$  yr). No lab run can reach the ocean floor at natural concentration.



design

**Choice: two tiers. 1. Realistic arm:** 1 mM, 20% seed, 3 years, 9 replicate measurements per time point. Reaches  $1.8\times10^{-6}$ , below the pond floor. This asks directly whether amplification happens at meteoritic concentrations. 2. **Accelerated arm:** 10 mM and 0.1 M for 1 year. These are **not natural conditions**. The arm exists to find any amplifier at all and to measure how the rate of ee growth scales with concentration (the rate law). The model needs that scaling to extrapolate to 1  $\mu$ M. If ee grows faster at 0.1 M than at 10 mM by exactly 10 $\times$ , k2 is a true second-order constant and the extrapolation is fair.

Arms (chemistries to test)

Chosen from the scorecard for prebiotic inputs (R4) and biological products (R7):

| arm | system                                                                            | why                                                                 | source                      |
|-----|-----------------------------------------------------------------------------------|---------------------------------------------------------------------|-----------------------------|
| A   | amino acids + carbonyl sulfide (volcanic gas) → peptides, in water                | prebiotic peptide formation; peptides can prefer same-hand partners | Leman, Orgel & Ghadiri 2004 |
| B   | amino acids + hydroxy acids, <b>wet-dry cycling</b> (e.g. daily)                  | pond-like driving; makes polypeptides                               | Forsythe et al. 2015        |
| C   | prebiotic cysteine peptides that <b>catalyse peptide joining</b> in neutral water | the closest known thing to self-copying in prebiotic water          | Foden et al. 2020           |

| arm | system                                                                                                                                      | why                                                                                                                                                                    | source                                                            |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| D   | RNA precursor (RAO) on <b>magnetite</b> , then crystallization                                                                              | non-autocatalytic route that already breaks symmetry                                                                                                                   | Ozturk et al. 2023                                                |
| E   | peptide-catalysed <b>transamination</b> (pyruvate + pyridoxamine → alanine) coupled to <b>peptide ligation</b> , run <b>fed (slow flow)</b> | the one proposed prebiotic network predicted to break symmetry on its own, built from two measured reactions; the model says it must be fed, not closed (review.py R6) | Yu et al. 2024; Deng, Yu & Blackmond 2024; Higgs & Blackmond 2025 |
| +   | <b>Positive control:</b> Viedma grinding of a known conglomerate                                                                            | proves the pipeline detects real amplification                                                                                                                         | Viedma 2005                                                       |

Arms E and C are the priority. E is the only prebiotic network with a published prediction of symmetry breaking (Higgs & Blackmond 2025), and its transamination step has a measured rate near  $10^{-4}$  /M/s at 0.1 M (our estimate from Yu et al. 2024, Table 2), well above the 1-year detection floor at 1 mM in the table above. Run E in a slowly fed vessel; a closed vial is predicted to return to 50/50. C is the only other catalytic loop in prebiotic water.

## Conditions per arm

For each arm, three seeds × 3 replicates: - **+20% L seed** - **+20% D seed** (the mirror control, the most important control) - **0% (racemic)**

Plus three negative controls (3 replicates each): no activator / no cycling; sterile-filtered; and the seed amino acid alone (to measure plain racemization). That’s 9 + 9 = 18 vials per arm, plus the positive control. Temperature 25 °C, with an optional 40 °C set to speed things up. Racemization is then negligible over a year for most amino acids, and the “seed alone” vials measure it directly.

## Sampling

Days 0, 3, 7, 14, 30, 60, 90, 120, 180, 270, 365, then every 6 months to 3 years for the realistic arm (roughly evenly spaced on a log scale). Each sample is measured 9 times. Measure ee of the monomers (and of short peptides where possible) by **chiral GC-MS or HPLC** on derivatized samples. Keep analysts **blind** to vial labels.

## Decision rule (fixed in advance)

Call an arm an **amplifier** only if **all four** hold: 1. |ee| rises by > 3.7% absolute over its start in both seeded sets ( $3\sqrt{2} \times 2.6\%/\sqrt{9}$ ). Use a smaller threshold if the lab’s measured precision is better. 2. The **L-seeded and D-seeded vials change by equal and opposite amounts** (within noise). 3. The racemic vials stay at 0 within noise. 4. Sterile and no-activator controls show no change.

Then fit  $ee(t) = e0 \cdot \exp(k \cdot t)$  to get k, and  $k2 = k / c$ . Compare k2 with the floors ( $3 \times 10^{-6}$  pond at 1 mM,  $3 \times 10^{-9}$  ocean at 1 μM), using the accelerated arm’s rate law to extrapolate.

# Contamination: the main way this goes wrong

Biology is left-handed, so any microbe, skin trace or reagent impurity adds **L**. That would look like “amplification” in the L-seeded vials. Safeguards: - **The D-seed mirror arm**: real amplification is symmetric; contamination pushes both sets toward L. - Sterile filtration, sealed vials, and a vial with no amino acid (it should show zero amino acids). - If only the L arm “amplifies”, report it as contamination, not a result.

## What each outcome means

| result                                                      | meaning for the project                                                                                                                                                              |
|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| An arm amplifies at $k_2 \geq$ pond floor                   | First prebiotic, water-based amplifier; pond seeding by meteorites becomes a complete story                                                                                          |
| An arm amplifies but slower than the floor                  | Real, but too slow to matter in a pond; ocean-scale only                                                                                                                             |
| Nothing amplifies at 1 mM in 3 years, or at 0.1 M in 1 year | Rules out these chemistries down to $k_2 \approx 2 \times 10^{-6}$ (realistic) or $5 \times 10^{-8}$ (accelerated). That’s still a publishable upper limit, and it tightens the spec |
| Only the positive control amplifies                         | Pipeline works; the gap in the scorecard is real                                                                                                                                     |

## Limits and practical notes

- This needs a chemistry lab with chiral GC-MS or HPLC. It’s a proposal for a mentor or a summer lab, not a home experiment. Local groups with chiral GC-MS (suggested by J. Dworkin): NASA Ames (G. Cooper, G. Chaban); San José State (A. Rios). Carbonyl sulfide is a toxic gas and needs fume-hood handling (arm A).
- The 2.6% noise figure comes from published repeat measurements. The lab’s own precision sets the real threshold; the smallest detectable  $k_2$  grows roughly in proportion to the noise.
- A null result limits only these specific chemistries, not all possible ones.
- The rate model is the single-exponential early phase from `sim.py`. Late saturation isn’t used for the fit.

## References

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